

11

1

Udemy course
ML, NLP, DL

Course Name is SAT

Set - 22
Inferential
Statistics

lec - 8 Bayes Theorem

lec - 9 Confidence
Interval And Margin
of Error

Bayes Theorem

Bayesian statistics is
an approach to data
analysis & parameter
estimation based on
Bayes Theorem

Bayes's Theorem

Probability

Independent
Events

Dependent
Events

① No explanation about dependent & Independent

① Independent Events

Ex:- Rolling a dice
{ 1, 2, 3, 4, 5, 6 }

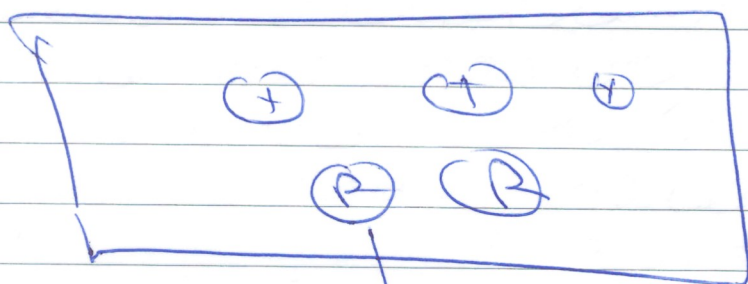
$$Pr(1) = \frac{1}{6} \quad Pr(2) = \frac{1}{6} \dots$$

Rolling a coin

$$Pr(H) = 0.5$$

$$Pr(T) = 0.5$$

② Dependent Events



Probability of Red

$$Pr(R) = \frac{2}{5}$$

After taking out a ball
now we want
 $Pr(\text{yellow now})$

(3)

$$Pr(+|blue) = \frac{3}{4}$$

→ The no. of marbles
 $\Rightarrow 5 - 1 = 4$

Now 4 marbles are
there

Hence now the probability
 $= \frac{3}{4} P.$

Now,
 $Pr(R \text{ and } +)$

→ It means that
first Red is taken
→ then + is taken.

$$Pr(R \text{ and } +) = Pr(R) + P(+|R)$$

condition
↑
Red

$$= \frac{2}{5} \times \frac{3}{4} = \frac{6}{20}$$
$$= \frac{3}{10} P.$$

$$Pr(A \text{ and } B) = Pr(B \text{ and } A)$$

$$Pr(A) * Pr(B|A) = Pr(B) * Pr(A|B)$$

$$Pr(B|A) = \frac{Pr(B) * Pr(A|B)}{Pr(A)}$$

→ This is
Bayes Theorem

A, B are basically
the events.

$Pr(A|B)$ = Prob. of A, given
B is true

$Pr(B|A)$ = Prob of B given
A is true

$Pr(A), Pr(B)$ = Independent
probabilities
of A and B

(5)

☆

Suppose a dataset is given

Size of House

x_1

No. of Rooms

x_2

Location etc

x_3 &

Independent Features

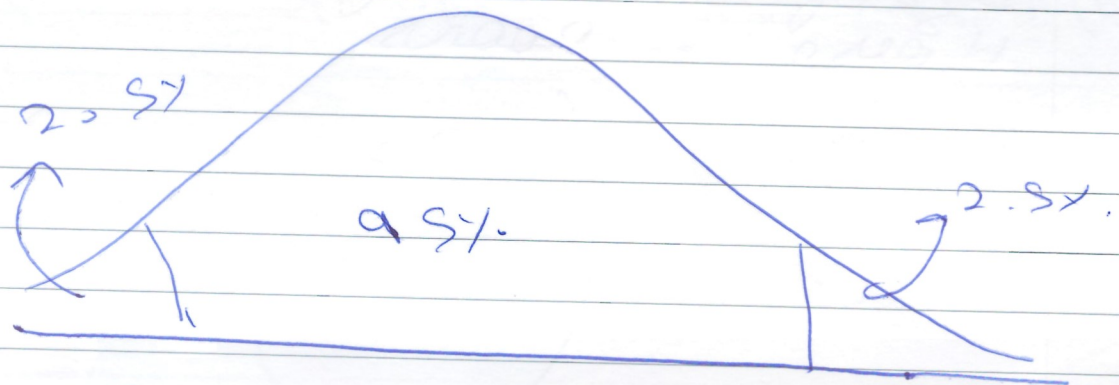
Dependent (Response) and Feature

The required eqn is -

$$P\left(\frac{y}{x_1, x_2, x_3}\right) = \frac{P(y) * P(x_1, x_2, x_3 | y)}{P(x_1, x_2, x_3)}$$

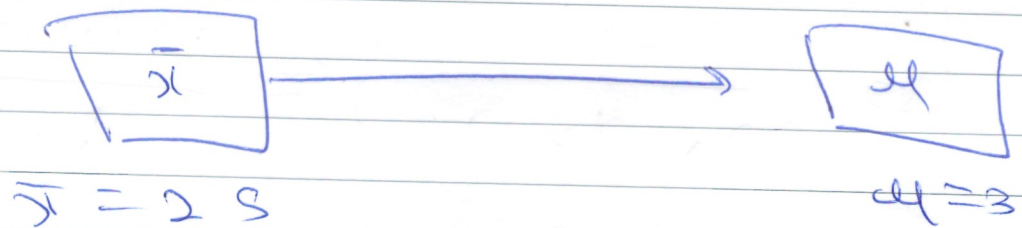
→ This is Bayes Theorem

Let α :- Confidence Interval $\rightarrow$ Margin of Error



Confidence Interval = 95%

Point Estimate



Confidence Interval

Point Estimate $\pm$ Margin of Error

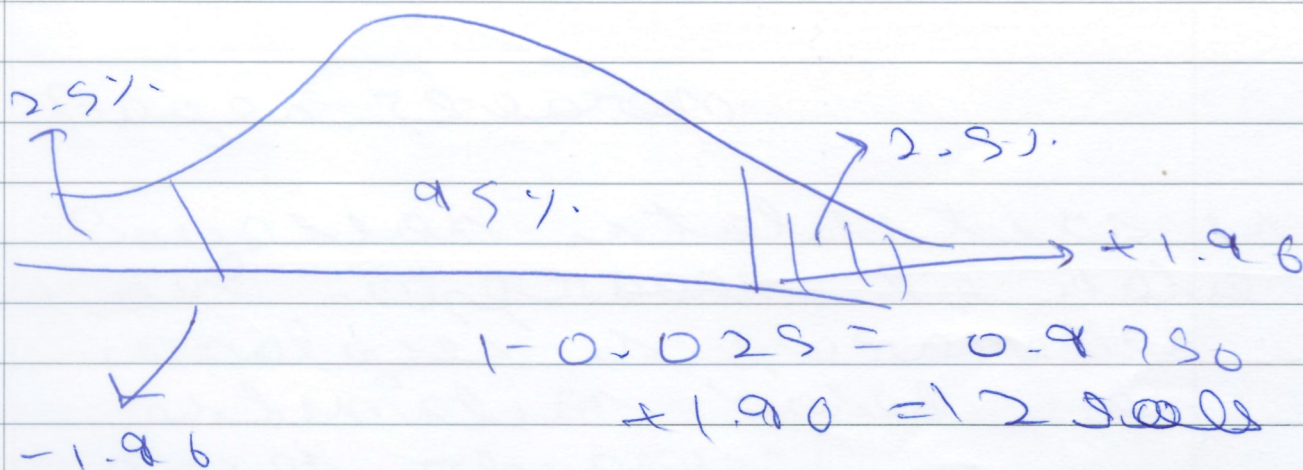
$$2 \text{ test} \Rightarrow \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Ex: on the vertical section
of CAT axis, the
std deviation is known
to be 100. A sample
of 30 test tubes
have a mean of 520

construct a 95% CI about
mean?

Soln $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$\alpha = 0.05$



Lower CI = $520 - 1.96 \frac{100}{\sqrt{30}}$
 $= 480.8$

Upper CI = $520 + (1.96) \frac{100}{\sqrt{30}}$
 $= 559.2$

conclusion:- I am 95%
confident about ~~mean~~
that the mean
CAT score is b/w
480-8 and 599-2